

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO2-AT2-Scenario Based

View Only

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QUESTIONS

Q1

Smart Hospital Patient
Registration System
20 marks

Q2

Online Library Book
Management System
20 marks

2/2 answered

Q2

Online Library Book Management
System20
marks

Scenario

Online Library Book Management System

A university library is planning to automate its book management process. The librarian wants a Java application that stores book details, prevents duplicate book entries, allows students to search for books quickly, and generates category-wise reports. The application should use Java String handling and Java Collection Framework to efficiently manage the library records.

Requirements

Each book contains the following details:

- Book ID
- Book Title
- Author Name
- Category
- Publisher

Use:

- ArrayList to store all book objects.
- HashMap to store Book ID as the key for quick searching.
- HashSet to maintain unique author names.

Menu Driven Program

1. Add Book
2. Search Book by ID
3. Display All Books
4. Display Unique Authors
5. Category-wise Book Count
6. Exit

Requirement 1 – Add Book (4 Marks)

Accept

Book ID

Book Title

Author Name

Category

Publisher

Validate the following:

Book ID must be unique.

Remove leading and trailing spaces.

Book Title should be stored in Title Case.

Category should be converted to uppercase.

Publisher name should not be empty.

Store the record in both

ArrayList

HashMap

Requirement 2 – Search Book (3 Marks)

Accept Book ID.

If found, display

Book ID

Book Title

Author

Category

Publisher

Otherwise,

Book Not Available

Requirement 3 – Display All Books (4 Marks)

Display all records in tabular format.

Example

ID TITLE AUTHOR CATEGORY

B101 Java Programming Herbert Schildt
PROGRAMMING

B102 Data Structures Mark Allen COMPUTER
SCIENCE

Requirement 4 – Display Unique Authors (3 Marks)

Display all distinct author names using HashSet.

Example

Unique Authors

Herbert Schildt

James Gosling

Yashavant Kanetkar

Mark Allen

Requirement 5 – Category-wise Book Count (4
Marks)

Display the number of books available in each
category using

HashMap<String,Integer>

Example

PROGRAMMING : 12

DATABASE : 8

NETWORKING : 6

Requirement 6 – String Operations (2 Marks)

For every book,

Display

□Length of Book Title

□Number of words in the title

□Category in uppercase

Example

Input

Java Programming Language

Output

Length : 25

Words : 3

Category : PROGRAMMING

Sample Input

1

B101

Java Programming

Herbert Schildt

Programming

McGraw Hill

1

B102

Data Structures

Mark Allen

Computer Science

Pearson

3

4

5

2

B101

6

Sample Output

Book Added Successfully.

Book Added Successfully.

Book ID Title Author

B101 Java Programming Herbert Schildt

B102 Data Structures Mark Allen

Unique Authors

Herbert Schildt

Mark Allen

Category-wise Count

PROGRAMMING : 1

COMPUTER SCIENCE : 1

Book Details

Book ID : B101

Title : Java Programming

Author : Herbert Schildt

Category : PROGRAMMING

Publisher : McGraw Hill

Constraints

Maximum books = 500

Duplicate Book IDs are not allowed.

Arrays are not allowed.

Use only Java Collections.

Book search should be performed using HashMap.

Use HashSet for maintaining unique authors.

Your Code

```
1 import java.util.*;
2 public class Main {
3     static ArrayList<Book> books = new Ar
4     static HashMap<String, Book> bookMap
5     static HashSet<String> authors = new
6     public static void main(String[] args
7         Scanner sc = new Scanner(System.i
8         while(sc.hasNextLine()) {
9             String input = sc.nextLine().
10             if(input.isEmpty()) continue;
11             int choice = Integer.parseInt
12             switch (choice) {
13                 case 1:
14                     String id = sc.nextLi
15                     if (bookMap.containsKey
16                         System.out.printl
17                         break;
18                 }
19             String title = sc.nex
```

```
20         String author = sc.ne
21         String category = sc.
22         String publisher = sc
23         if (publisher.isEmpty
24             System.out.printl
25             break;
26     }
27
28     Book book = new Book(
29         id,
30         title,
31         author,
32         category,
33         publisher
34     );
35     books.add(book);
36     bookMap.put(id, book)
37     authors.add(book.auth
38     System.out.println("E
39     break;
40 case 2:
41     String searchId = sc.
42     if(bookMap.containsKe
43         Book b = bookMap.
44         System.out.printl
45         System.out.printl
46         System.out.printl
47         System.out.printl
48         System.out.printl
49         System.out.printl
50     }
51     else {
52         System.out.printl
53     }
54     break;
55 case 3:
56     System.out.println("-
57     System.out.println("E
58     System.out.println("-
59     for (Book b : books) {
60
61         b.id + "
62         b.title +
63         b.author
64     );
65     }
66
67     break;
```

```
68         case 4:
69             System.out.println("U
70             for(String a : author
71                 System.out.printl
72             }
73             break;
74         case 5:
75             System.out.println("C
76             HashMap<String, Integ
77                 new HashMap<>
78             for(Book b : books){
79                 if(count.contains
80                     int oldCount
81                     count.put(b.c
82                 } else {
83                     count.put(b.c
84                 }
85             }
86             for(Map.Entry<String,
87                 count.entrySet
88
89                 System.out.printl
90                     entry.get
91                     entry.get
92             );
93         }
94
95         break;
96         default:
97             return;
98     }
99 }
100 }
101
102 static class Book {
103     String id;
104     String title;
105     String author;
106     String category;
107     String publisher;
108     Book(String id,
109         String title,
110         String author,
111         String category,
112         String publisher) {
113         this.id = id.trim();
114         this.title = toTitleCase(titl
115         this.author = author.trim();
```

```
116         this.category = category.trim
117         this.publisher = publisher.tr
118     }
119 }
120 static String toTitleCase(String str)
121     String result = "";
122     boolean newWord = true;
123     for (int i = 0; i < str.length();
124         char ch = str.charAt(i);
125         if(Character.isWhitespace(ch)
126             result = result + ch;
127             newWord = true;
128         }
129         else {
130             if(newWord) {
131                 result = result +
132                     Character.toU
133                 newWord = false;
134             }
135             else {
136                 result = result + Cha
137             }
138         }
139     }
140     return result.trim();
141 }
142 }
```



Input

stdin input

Output

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



0.00 / 20 marks

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